

Muhammad Taha

AI Engineer | Computer Vision, OCR, LLM/RAG, Backend Automation
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Summary

AI engineer building production-oriented systems for video understanding, document processing, LLM workflows, and backend automation. I focus on the engineering that makes AI useful after the demo: data quality, evaluation, failure modes, interfaces, deployment, logging, and repeatable operations.

Recent work includes a basketball made-basket clip finder for a US sports client, a document preprocessing pipeline for scanned county records, a local document de-identification prototype, LLM/RAG systems, guarded data-analysis tools, and full-stack review workflows.

Professional Experience

Basketball Video Analysis System - Freelance AI Engineer

Client project | Mar 2026 - Aug 2026

- Built a computer-vision workflow that scans basketball game footage and exports made-basket highlight clips with timestamped 20-second windows.
- Combined hoop/ball detection, trajectory checks, and a verification stage to reduce false positives while keeping recall high for real makes.
- Delivered a dashboard workflow with queue handling, time-range selection, live logs, Windows launch scripts, and a false-positive review panel for fast client cleanup.
- Received public 5.0 client feedback for speed, flexibility, and quality of work.

Document OCR Preprocessing Pipeline - Client AI Engineer

US records-processing environment | Active client work

- Built a Python/OpenCV preprocessing pipeline for scanned county property records, preserving Year/Month/Date folder structure and multi-page TIFF page order.
- Implemented orientation correction, deskewing, artifact cleanup, denoising, illumination normalization, contrast improvement, and adaptive thresholding.
- Added configuration profiles, resumable execution, failure tracking, runtime logs, worker controls, and Windows operations documentation for approved client infrastructure.

Gohar Textile Mills Pvt Ltd. - AI Engineer Intern

Faisalabad | Jun 2026 - Aug 2026

- Built local AI and data workflows around SAP production logs, sales matrices, and executive reporting needs.
- Developed Python automation pipelines for shift operations, document handling, and data-entry reduction.
- Implemented sales forecasting and backtesting workflows using SAP OData, XGBoost, Prophet, seasonal baselines, MAPE scoring, and Streamlit dashboards.

AWS Cloud Club FAST-NUCES - Co-Lead, Development & Projects

Peshawar | Oct 2025 - Present

- Helped lead cloud-focused student projects, workshops, and mentoring around AWS architecture, deployment, and production habits.
- Guided teams on EC2, Lambda, serverless workflows, CI/CD, and practical project delivery.

Selected Projects

MCP Data Analyst Python, FastAPI, PostgreSQL, MCP, Docker

Built a guarded PostgreSQL analysis service exposing query, statistics, and chart workflows through MCP tools and REST endpoints.

Used SQL AST validation, read-only transactions, statement timeouts, row limits, and query-cost controls around generated SQL.

Rabt / Context Optimization Python, AST parsing, NetworkX, Gemini

Built graph-based code-context tooling that extracts smaller relevant subgraphs for code-aware LLM workflows.

Focused on context reduction, dependency understanding, and inspectable memory/context construction.

Local Document De-ID Prototype Python, DOCX/PDF, OCR, Tesseract, Poppler

Built a local review workflow for sensitive Word/PDF text with rule-based detection, manual redaction, mandatory approval, verified DOCX export, and loopback-only UI boundaries.

QR Payment Verification React, Express, MongoDB, Gemini Vision

Built a receipt-review workflow with OCR extraction, duplicate detection, weighted risk signals, async upload status, and an admin review interface.

Education

Bachelor of Science, Computer Science

National University of Computer and Emerging Sciences (FAST-NUCES) | Aug 2023 - Aug 2027 expected

- Relevant coursework: machine learning, data structures and algorithms, advanced databases, compiler construction, operating systems, and computer networks.

Core Skills

AI engineering: computer vision, OCR/document AI, document privacy workflows, LLM applications, RAG, MCP integrations, prompt engineering, evaluation, failure analysis, and human-in-the-loop workflows.

Backend and data: Python, FastAPI, Flask, REST APIs, PostgreSQL, SQL, MongoDB, Pandas, NumPy, Parquet, data pipelines, validation, and audit-friendly processing.

Computer vision: OpenCV, YOLOv8, EfficientNet, image preprocessing, object detection, video analysis, OCR readiness, perceptual hashing, and risk scoring.

Product delivery: dashboards, client workflows, Docker, Git, Linux, Windows operations, FFmpeg, AWS EC2, Firebase, CI/CD, logging, resumable jobs, and deployment documentation.

Frontend and apps: React, Next.js, TypeScript, JavaScript, Streamlit, Flutter, Dart, Firebase Auth, and Firestore.

How I Work

I do not start with a model choice. I start with the workflow, data quality, constraints, review process, and failure cost. Then I choose the simplest system that can reliably solve the problem: deterministic processing, computer vision, RAG, LLM tooling, or a hybrid.

I care about production details that make clients trust the system: clear setup, repeatable runs, logs, resume behavior, safe defaults, review screens, documented limitations, and maintainable code.

Certifications

Model Context Protocol | AI Fluency: Framework & Foundations | Google Cybersecurity | Claude Code in Action